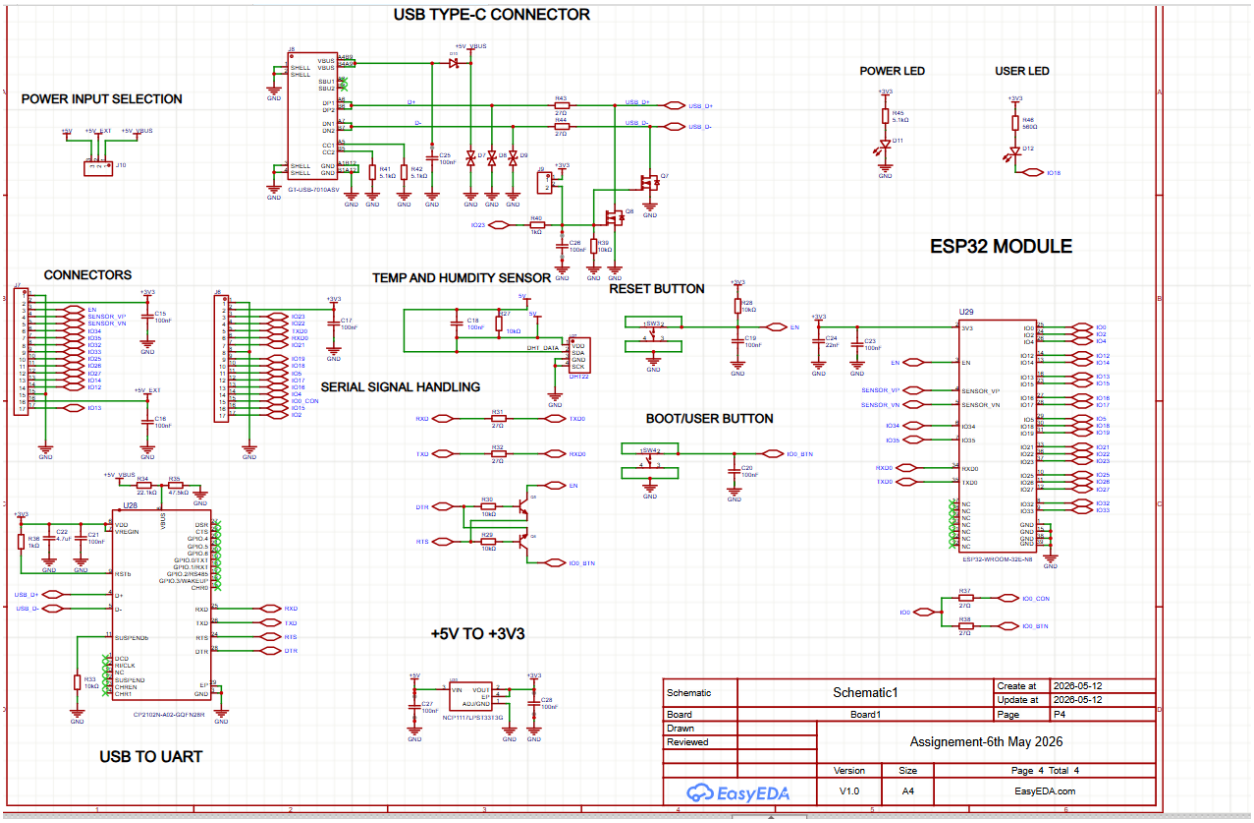
PART B

Number 2



Table

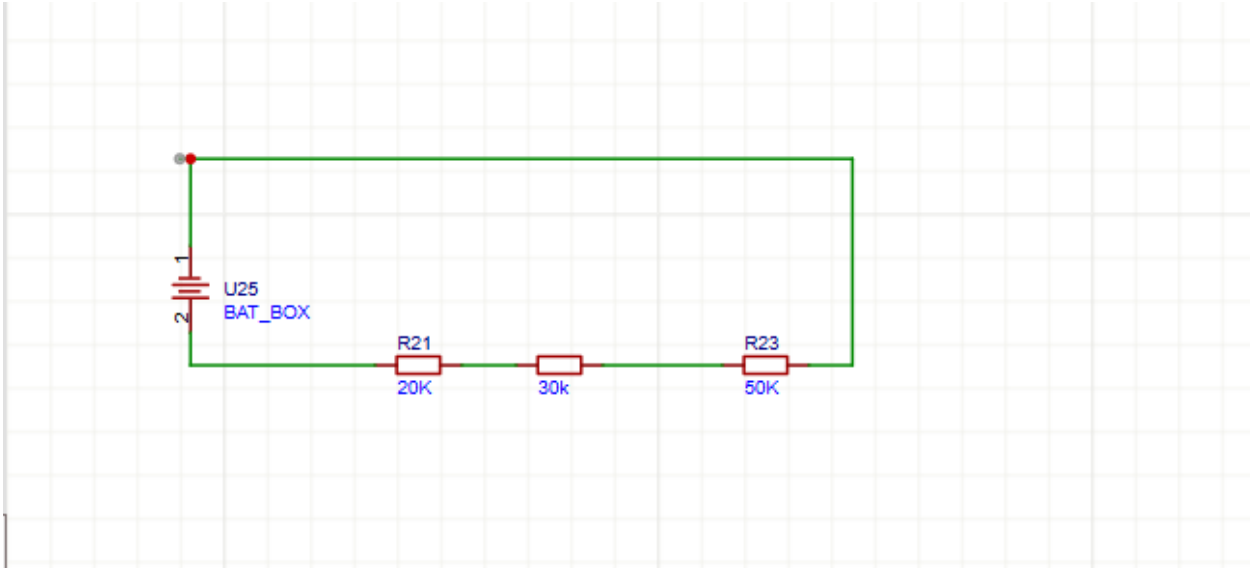
Number	Reference	Component	Value/Model	Quantity
1	U29	Microcontroller Module	ESP32-WROOM-32E-N8	1
2	U28	USB to UART Bridge	CP2102N-A02-GQFN28R	1
3	U30	Voltage Regulator (5V→3.3V)	NCP1117PST33T3G	1
4	DHT22	Temperature & Humidity Sensor	DHT22	1
5	J10	Power Input Selection Jumper	X1311WV-03J-C40D24	1
6	J8	USB Type-C Connector	GT-USB-7010ASV	1
7	J9	Jumper	DZ127R-11-02-25	1

8	1SW32	Reset Button	Tactile Push Button	1
9	1SW42	Boot/User Button	Tactile Push Button	1
10	D11	Power LED	LED	1
11	D12	User LED	LED	1
12	Q7	Transistor	2N7002LT1G	1
13	Q8	Transistor	2N7002LT1G	1
14	R45	Resistor (Power LED)	5.1kΩ	1
15	R46	Resistor (User LED)	500Ω	1
16	R27	Pull-up Resistor (DHT22)	5V	1
17	R28	Pull-up Resistor (Reset)	10kΩ	1
18	R29	Resistor (Serial)	10kΩ	1
19	R30	Resistor (Serial)	10kΩ	1
20	R31	Resistor (Serial Signal)	27Ω	1
21	R32	Resistor (Serial Signal)	27Ω	1
22	R33	Resistor (UART)	10kΩ	1
23	R34	Resistor (USB)	22.1kΩ	1
24	R35	Resistor (USB)	47.5kΩ	1
25	R37	Resistor (IO)	27Ω	1
26	R38	Resistor (IO)	1kΩ	1
27	R39	Resistor	10kΩ	1
28	R40	Resistor	1kΩ	1
29	R41	Resistor (USB)	5.1kΩ	1
30	R42	Resistor (USB)	5.1kΩ	1
31	R43	Resistor (USB D+)	27Ω	1
32	R44	Resistor (USB D-)	27Ω	1
33	C17	Capacitor (DHT22 area)	100nF	1

34	C18	Capacitor (DHT22 area)	100nF	1
35	C19	Capacitor (Reset Button)	100nF	1
36	C20	Capacitor (Boot Button)	100nF	1
37	C21	Capacitor (UART)	100nF	1
38	C22	Capacitor (UART)	4.7µF	1
39	C23	Decoupling Capacitor (ESP32)	100nF	1
40	C24	Decoupling Capacitor (ESP32)	22nF	1
41	C25	Capacitor (USB)	100nF	1
42	C27	Filter Capacitor (+5V to +3V3)	100nF	1
43	C28	Filter Capacitor (+5V to +3V3)	100nF	1
44	C15	Connector Capacitor	100nF	1
45	C16	Connector Capacitor	100nF	1
46	D7-D9	Diodes (USB Protection)	LESD5D5.0CT1G-ES	4

Number 1

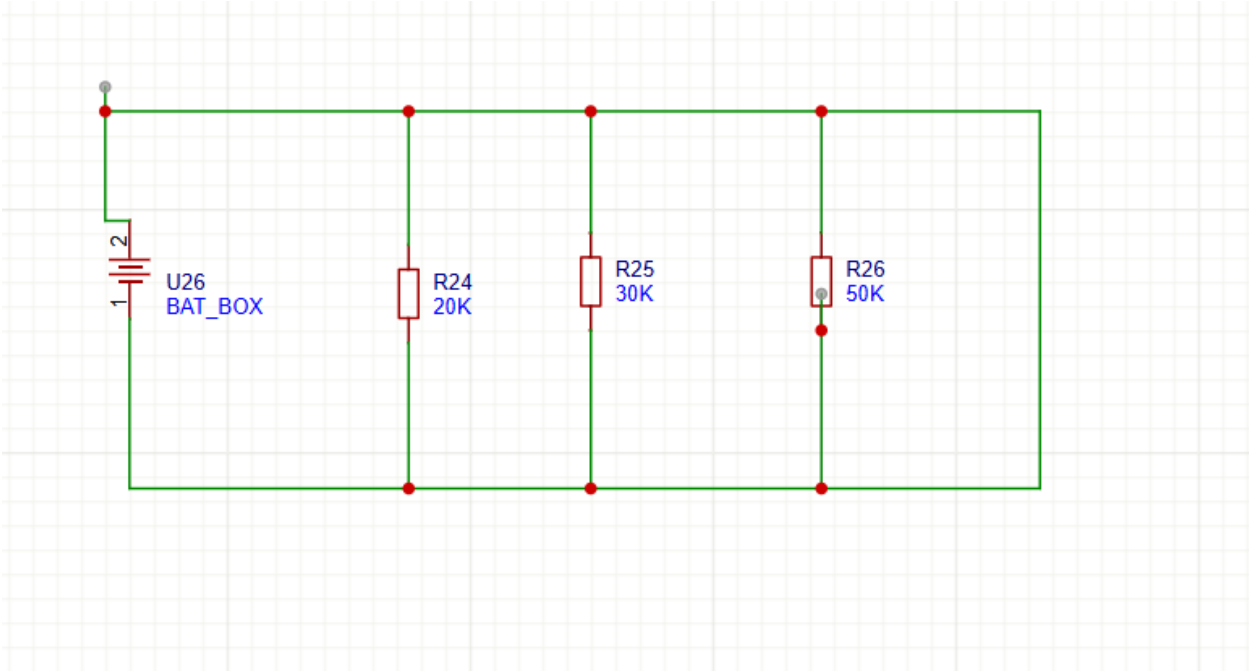
Circuit A



Table

Number	Reference	Component	Value	Quantity
1	U25	DC Power Supply	125V	1
2	R21	Resistor	20Ω	1
3	R22	Resistor	30Ω	1
4	R23	Resistor	50Ω	1

Circuit B



Table

Number	Reference	Component	Value	Quantity
1	U26	DC Power Supply	125V	1
2	R24	Resistor	20Ω	1
3	R25	Resistor	30Ω	1
4	R26	Resistor	50Ω	1